5. Vibration

Introduction

This chapter discusses the characteristics of vibration, the theory of vibration, and the effects of vibration on man. In addition, it discusses the methods of controlling vibration; in particular, how to select isolators that isolate a disturbing force from a radiating surface.

Definition of Vibration

In general, vibration can be described as an oscillary motion in a system about an equilibrium position produced by a disturbing force. The three key words are motion, system, and disturbing force.

The motion may be flexorial, torsional, compressional, or more complex. There may be one or more modes of vibration at different frequencies for each type of motion. The disturbing force may be periodic or nonperiodic. Periodic disturbing forces are produced mechanically by unbalanced, misaligned, loose, or eccentric parts of rotating machinery; for example, bad gears or bent shafts. Forces are also produced hydraulically and aerodynamically; for example, by pump propellers, centrifugal compressors, or fans. In either case, the frequency spectrum occurs at the basic rotational speed or some multiple of it. Nonperiodic forces are produced by sliding or rolling parts, turbulent fields, or jet discharges; for example, bad bearings or cavitation. The resulting frequency spectrum is determined by the interaction of the inherent properties of the system with the disturbing force. It is not necessarily related to the rotational speed for nonperiodic forces.

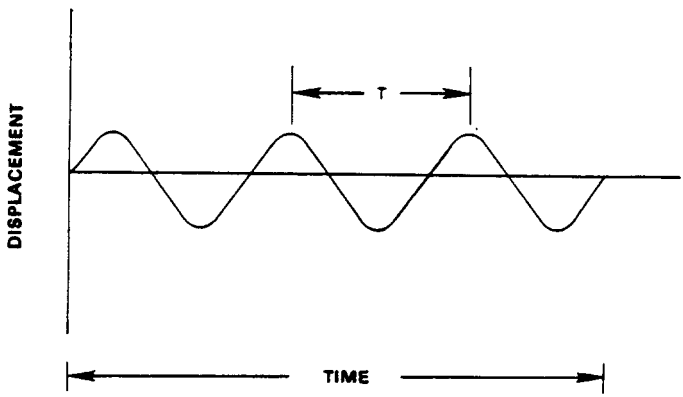
The system referred to in the definition of vibration may be gaseous, liquid, or solid. When the system is air (gaseous) and the motion involves vibration of air particles in the frequency range of 20,000 Hz, sound is produced. However, usually the word "vibration" is used to describe motions of the structures found in machinery, bridges, or battleships and is classed as solid-borne or mechanical vibration.

Periodic Vibration

Vibration is considered periodic if the oscillating motion of a particle around the position of equilibrium repeats itself exactly after some period of time. The simplest form of periodic vibration is called "pure harmonic motion" which can be represented by a sinusoidal curve as a function of time. Such a relationship is illustrated in Figure 4.5.1.

Figure 4.5.1

Periodic vibration.



The motion of any particle can be characterized at any time by (1) displacement from equilibrium position, (2) the velocity or rate of change in displacement, or (3) acceleration, or the rate of change in velocity. For pure harmonic motion, the three characteristics of motion are related mathematically.

The instantaneous displacement of a particle from its reference position under the influence of harmonic motion can be described mathematically as follows:

$$\begin{aligned} s &= S \sin(2\pi t/T) \\ &= S \sin (2\pi f) \\ &= S \sin \omega t \end{aligned}$$

where s denotes instantaneous displacement, S denotes maximum displacement, t denotes time, T denotes period of vibration, f denotes frequency, and ω denotes angular frequency ($2\pi f$).

Of the possible vibration measurements (displacement, velocity, and acceleration), displacement is probably the easiest to understand. Displacement can be measured directly only if the rate of motion, i.e., frequency of vibration, is low enough.

In many practical problems, displacement is not the most important property of vibration. For example, experience has shown that velocity of a vibrating part is the best single criterion for use in preventive maintenance of rotating machinery.

Although peak-to-peak displacement measurements have been widely used for this purpose, it is necessary to establish the relationship between the limits for displacement and rotational speed for each machine. Since the velocity of

a moving particle is a change in displacement with respect to time, velocity can be mathematically defined as

$$v = ds/dt$$

$$= \omega S \cos(\omega t)$$

$$= v_{max} \cos(\omega t)$$

$$= v_{max} \sin(\omega t + \pi/2)$$

where v denotes instantaneous velocity and v_{max} denotes maximum velocity.

In many cases of vibration, especially when mechanical failure is a consideration, actual forces set up in the vibrating parts are critical factors. Since the acceleration of a particle is proportional to these applied forces and since equal and opposite reaction forces result, particles in vibrating structures exert forces that are a function of the masses and acceleration of the vibrating parts on the total structure. Thus, acceleration measurements are another means by which the motion of vibrating particles can be characterized.

The instantaneous acceleration--the time rate of change of velocity of a particle in pure harmonic motion--can be described as follows:

$$a = dv/dt$$

$$= d^2s/dt^2$$

$$= -\omega^2 S \sin(\omega t)$$

$$= A \sin(\omega t + \pi)$$

where "a" denotes instantaneous acceleration and "A" denotes maximum acceleration.

The expressions given above have utilized peak values. This approach is quite useful in the consideration of pure harmonic vibration. However, in the situation of more complex vibration, peak values are not that useful. What needs to be known is the behavior of the vibration over its history. Peak values do not take into account the previous history of the vibration. One descriptive quality that does take history into account is the average absolute value.

$$S_{average} = 1/T \int_0^T |s| \, dt$$

Although the above expression does take the time history of the vibration into account (over one period), it has very limited practical usefulness. The root mean square value is much more useful.

$$S_{rms} = [1/T \int_0^T s^2 dt]^{1/2}$$

For pure harmonic vibration, it can be shown that

$$S_{rms} = \pi/2 \sqrt{2}$$

Attempts have been made to introduce the concepts of displacement level, velocity level, and acceleration level. These are similar in concept to sound-pressure level, where displacement, velocity, and acceleration are expressed in decibels. Displacement level is defined as

$$\text{Displacement level} = 20 \text{ Log}_{10}(d/d_o)$$

where d_o denotes the reference quantity usually given as 10^{-11} meters or 39.4×10^{-11} inches.

Velocity level is given as

$$\text{Velocity level} = 20 \text{ Log}_{10}(v/v_o)$$

where v_o denotes the reference quantity usually given as 10^{-8} m/sec or 39.4×10^{-8} inches/sec.

The acceleration level is given as

$$\text{Acceleration level} = 20 \text{ Log}_{10}(a/a_o)$$

where a_o denotes the reference quantity usually given as 10^{-5} m/sec^2 or 39.4×10^{-5} inches/sec^2.

When to Use What

Vibration can be examined by looking at displacement, velocity, or acceleration. Although displacement of the moving part is easiest to understand, in many practical problems displacement is not the most important property.

A vibrating part will radiate sound in much the same way as a common loudspeaker does. In general, velocities of the radiating part (which corresponds to the cone of the loudspeaker) and the air next to the part will be the same; and if the distance from the front of the part to the back is large compared with one-half of the wavelength of the sound in air, the actual sound pressure in air will be proportional to the velocity of the vibration. Furthermore, the sound energy radiated by the vibrating surface is the product of the velocity squared and the resistance components of the air load. Under these conditions, it is the velocity of the vibrating part and not its displacement that is of greater importance.

In the above paragraph, it was shown that in the case where the vibrating part is large compared to the wavelength of the sound that velocity was of importance. However, in most machines this relationship does not hold, since relatively small parts are vibrating at relatively low frequencies. This situation can be compared to a small loudspeaker without a baffle (cone). At low frequencies, the air may be pumped back and forth from one side of the cone to the other with high velocity but without building up much pressure or radiating much sound energy because of the very low air load. Under these conditions, an acceleration measurement provides a better measure of the amount of noise radiated than does the velocity measurement.

Acceleration measurement has been extensively used to analyze the situation where vibration is severe enough to cause mechanical failure. In mechanical failure, the actual forces set up in the vibrating parts are important factors. The acceleration of a given mass is proportional to the applied force, and the reacting force will be equal but in the opposite direction. Members of a vibrating structure, therefore, exert forces on the total structure that are a function of the masses and the accelerations of the vibrating parts. Thus, acceleration measurements are the important measure when actual mechanical failure due to vibration is considered.

In summary, for a given sinusoidal displacement, velocity is proportional to frequency; and acceleration is proportional to the square of the frequency. The higher-frequency components of a vibration are progressively more important in velocity and acceleration measurements than in displacement readings.

Resonance

An object when struck will vibrate at its own frequency. This frequency depends upon the elasticity and shape of the object. Resonance occurs when successive impulses are applied to a vibrating object in time with its natural frequency. When an elastic object is struck with the sound wave of the same natural frequency, the wave will set the object into resonance--motion. If there are two matching tuning forks and one is set into motion, the other one can be set into motion. When tuning forks are placed a specified distance apart, the first tuning fork, when put into motion, will cause the other tuning fork to vibrate since they both are at the same frequency. What happens is this: The areas of compression give the prongs of the fork a tiny push. Since these pushes correspond to the natural frequency of the fork, they will successively increase the amplitude (displacement) of the vibration.

The effects of resonance can be extremely disastrous. In 1940, the Tacoma Narrows Bridge was destroyed by wind generating a resonance.

During the discussion on how to select isolators to control vibration, the term and concept of resonance will be used.

Vibration Measurement

A wide variety of systems can be used to measure vibration; mechanical systems, electrical systems, and optical systems such as stroboscopic

observation are examples. A stroboscopic system is discussed in Handbook of Noise Measurement, General Radio, page 224. The most common system used has three components: a vibration pickup, which transforms mechanical motion into an electrical signal; an amplifier, to enlarge the signal; and an analyzer, to measure the vibration in specific frequency ranges. Various types of each measuring device are available; e.g., pickups can measure displacement, velocity, or acceleration. The most common type of pickup is the accelerometer.

The problems involved in measuring vibration are determining what quantity to measure--displacement, velocity, or acceleration; determining the points at which the vibration pickup should be placed; determining how to fasten the pickup; and determining how to arrange the equipment in the field.

The quantities to be measured and some general rules of when to use what have already been discussed.

The correct pickup location is sometimes not obvious, and some exploration of the vibration position is required. Furthermore, the pickup must be correctly oriented, and this too sometimes requires exploration. The pickup must be oriented with respect to the direction of motion. In general, vibration instruments are most sensitive to vibrations in the direction perpendicular to the largest flat surface of the pickup.

The direction of maximum vibration at a point is often obvious from the vibrating structure; it is usually in the direction of least stiffness. However, this rule is sometimes misleading because of the many possible modes of vibration, some of which are perpendicular to the obvious direction of least stiffness. Such a mode can be strongly excited if close to the frequency of a component of the driving or disturbing force. Furthermore, the nature of the motion may favor one mode of vibration rather than another.

The measuring location is then related to the detection of the source of the disturbing force and to the identification of surfaces contributing significantly to airborne sound. Seemingly insignificant differences in location can result in significant differences in the observed reading. Usually readings are taken at more than one location and then compared. The comparison then helps to locate the significant sources. Rotating machinery provides a good example. Radial (perpendicular to the shaft), horizontal, and vertical readings on the bearing housing (as close as possible to the shaft) are compared to axial (parallel to the shaft) readings, based upon the knowledge that pickup response is directional.

Unbalance usually produces radial vibration at the basic rotational speed. Misalignment does the same, but is also associated with axial vibration at two or three times the basic speed. Looseness produces radial vibration at twice the basic rotational speed. Bad bearings produce high-pitched sounds unrelated to the basic speed. Gears, fans, impellers, etc., produce vibrations at tooth-on-blade passing frequencies, which equal rotational speed multiplied by the number of passes per rotation. More complex procedures can be used which are beyond the scope of this text. (See Handbook of Noise Measurement, General Radio, pages 207 to 215.)

Pickups can be fastened to the vibrating surface in a number of different ways. Pickups may be attached temporarily with magnets, putty, double-sided tape, and grease (jelly). Or, they may be attached permanently with bolts or epoxy cement. Permanent attachments may be necessary for maintenance tests. Care must be taken when fastening the pickup since the method of fastening may cause the vibrating surface to change its characteristics which would result in erroneous readings.

In some cases, for example, to make a quick check of vibration amplitude or to explore a vibration pattern, a hand-held pickup might be used. If the vibrating machine is massive and is vibrating with a significant amplitude, this technique can be useful for frequencies below about 1000 Hz. There are enough serious limitations to this technique, however, so that it should not generally be expected to yield accurate or highly reproducible results.

Other errors in measurement can be introduced by magnetic and electrical fields such as a ground loop-induced electrical hum. Thus, care must be taken to eliminate any mechanical vibration of the cables used in the field situation. Temperature can also affect vibration measurement, particularly measurement of very low frequency and amplitude.

The Effects of Vibration

Besides the disturbing force setting a surface into vibration and creating noise, vibration can cause other problems--problems that are not necessarily related to the noise produced by vibration.

The human body is an extremely complex physical and biological system. When looked upon as a mechanical system, it contains a number of linear and nonlinear systems, the mechanical properties of which differ from person to person. Biologically, and certainly physiologically, the system is by no means simple. On the basis of experimental studies, as well as documented reports of industrial experience, it is apparent that exposure of workers to vibration can result in profound effects on the human body mechanically, biologically, physiologically, and psychologically. Although not much is known about the effects of vibration on the human body, there is some literature available, mainly in European countries.

There are really two types of vibration effects on the human body, whole-body vibration, and segmental vibration. The human body is subjected to mechanical vibration just as any other system is. For example, when the body sits on a tractor seat, the body can vibrate just as the tractor seat does. This is called "whole-body vibration." "Segmental vibration" refers to the vibration of only parts of the body such as the hands using hand tools, the feet, etc. In segmental vibration, only the body part is free to vibrate while the rest of the body is firmly supported.

As a mechanical system, the body and even the body parts have their own natural frequencies. That is, the body and its parts will vibrate or be sent into resonance. Whole-body resonance occurs at 3 to 6 Hz and from 10 to 14 Hz. Body parts can also be sent into vibration. For example, the head and

shoulders will vibrate at 20 to 30 Hz; the eyeballs will vibrate from 60 to 90 Hz; and the lower jaw and skull will vibrate at 100 to 200 Hz.

The effects of whole-body vibration are not completely understood. When the whole body vibrates, there is increased consumption of oxygen which affects the pulmonary system and cardiac output. In addition to pulmonary and circulatory changes, whole-body vibration can cause tendon reflexes. It can also impair the ability to regulate posture. In addition, there is some evidence to suggest that whole-body vibration can affect the electrical activity of the brain. It can also affect visual acuity and the ability to perform different types of motor activities.

The effects of segmental vibration are various. There are four types of disorders that can result from segmental vibration.

1. Raynaud's phenomenon, sometimes called "white fingers."

2. Neuritis and degenerative alterations, particularly in the ulnary and axillary nerves; that is, the loss of touch, the loss of sensitivity to heat, and even paralysis.

3. Decalcification of the bones.

4. Muscle atrophy (tenosynovitis).

Raynaud's Syndrome. Raynaud's syndrome affects about one-half of the workers exposed to segmental vibration. Sometimes this syndrome is called "dead fingers" or "white fingers." In Raynaud's syndrome the circulation of the hand becomes impaired; and when exposed to cold, the fingers become white and void of sensation as though mildly frosted. The condition usually disappears when the fingers are warmed for some time, but a few cases have been sufficiently disabling that some workers have been forced to seek other types of work. In some instances, both hands have been affected. The syndrome has been observed in a number of occupations including the use of fairly light vibrating tools such as air hammers for scraping and chipping metal surfaces, stone cutting, lumbering, and cleaning departments of foundries where workers have a good deal of overtime work. Most commonly the frequency implicated in Raynaud's syndrome is between 40 and 125 Hz. Exposure time of a few months can cause the symptoms to appear. When exposure ends, symptoms will improve; however, complete recovery is rare. There are no standards available concerning Raynaud's disease or symptoms, and most of the studies concerning the disease have been performed in European countries. Table 4.5.1 presents some of the evidence that has been found in these European countries for segmental and whole-body vibration. In recent years there has been an increase in studies concerning vibration.

Control of Vibration

Before control techniques are discussed, it should be pointed out that at the present time there are no generally accepted safe vibration levels for whole body or segmental vibration. Some standards do exist, but these are based on studies that produce variable results. Consequently, no reliable

Table 4.5.1

European countries in which clinical evidence of overexposure to workers to vibration has been reported.

Industry	Vibration Type	Common Vibration Sources
Agriculture	Whole body	Tractor operation
Boiler making	Segmental	Pneumatic tools
Construction	Whole body Segmental	Heavy equipment vehicles, pneumatic drills, jackhammers, etc.
Diamond cutting	Segmental	Vibrating hand tools
Forestry	Whole body Segmental	Tractor operation. chain saws
Foundries	Segmental	Vibrating cleavers
Furniture manufacture	Segmental	Pneumatic chisels
Iron and steel	Segmental	Vibrating hand tools
Lumber	Segmental	Chain saws
Machine tools	Segmental	Vibrating hand tools
Mining	Whole body Segmental	Vehicle operators rock drills
Riveting	Segmental	Hand tools
Rubber	Segmental	Pneumatic stripping tools
Sheet metal	Segmental	Stamping equipment
Shipyards	Segmental	Pneumatic hand tools
Stone dressing	Segmental	Pneumatic hand tools
Textile	Segmental	Sewing machines, looms
Transportation (operators and passengers)	Whole body	Vehicle operation

Source: The Industrial Environment: Its Evaluation and Control

standards have been constructed. In addition, most authors point out that the available standards should be used with caution in the industrial setting, mainly because the standards that have been developed are for short-term exposures and, therefore, do not reflect the typical industrial setting. Also, because these standards were developed using physically fit individuals (pilots), they do not necessarily represent the typical industrial population.

Vibration can be controlled in the following ways:

1. By reducing the mechanical disturbance (driving force) causing the vibration.

2. By isolating the disturbing force from the radiating surface.

3. By reducing the response of the radiating surface.

In order to discuss how to reduce the mechanical driving force, it should be understood that there are different kinds of forces: rotational force, impact force, and sliding force. Rotational forces can be controlled by:

1. Balancing;
2. Alignment;
3. Reducing clearances;
4. Replacing worn parts;
5. Proper lubrication;
6. Reducing the speed of the machine.

Impact forces can be reduced by:

1. Having a smaller force over a longer period of time;
2. Using helical instead of spur gears;
3. Putting fiber gears between metal gears;
4. Cushioning applications to control impact of parts falling against chutes.

Sliding forces can be controlled by proper lubrication and reduction of friction.

Another way of controlling vibration is to isolate the driving force from the vibrating surface. For example, if a vibratory object like an unbalanced machine is set on an ordinary wood table, the vibration will radiate to the table when the machine is started; and the table will begin to vibrate. The vibration being transmitted from the machine to the table can be reduced by somehow isolating the machine from the table. Isolators such as rubber washers can be put under the machine. The rubber washers will stop the machine from transmitting force to the table and, thus, stop the table from vibrating.

Installation and selection of isolators can be a difficult job because improper application of isolators can make the problem worse and may further increase the force being transmitted to the vibratory surface. Further, isolation is a difficult job because vibration can exist in six modes. For

example, there are three linear modes: horizontal, vertical, and transverse. The other three are known as rotational modes, and they appear as rocking actions around the center of gravity of the load that vibrates back and forth around the x, y, and z axis.

When developing an isolation system, there are several things that must be known--the natural frequency of the system and the frequency of the driving force. In the linear modes, the natural frequency can be computed using the following:

$$f_n = (1/2\pi)\sqrt{Kg/W}$$

where

f_n denotes natural frequency (Hz)
K denotes stiffness of the isolator (lbs/in)
g denotes acceleration in gravity, 386 inches (32.2 x 12) per second^2
W denotes weight of the system per mount (lbs)

An alternate method, the static deflection method, could be

$$f_n = 3.13 \times \sqrt{1/S_{st}}$$

where
f_n denotes natural frequency (Hz)
S_{st} denotes static deflection of the mount (inches), usually provided by the manufacturer of the isolator

For rotational modes, the following equation can be used:

$$f_n = (1/2\Pi)\sqrt{Kr/i} = 0.167\sqrt{Kr/i}$$

where
f_n denotes the natural frequency (Hz)
Kr denotes the rotational stiffness of the mount (in-lb/radian of angular displacement about a given axis)
i denotes mass moment of inertia of supporting load about a given axis through the center of gravity (pounds/inches^2)

The natural frequency can change if any of the following factors change: weight of the system, stiffness of the mount, moment of inertia, and rotational motion of the mount. These changes will change the static deflection and, therefore, the natural frequency. In most cases, changing the stiffness of the mount is the easiest way to alter the rotational frequency.

The second thing item must be known in order to design a good isolation system is the frequency of the driving force. If there are many frequencies involved, the lowest frequency of the system must be used.

Transmissibility is also a factor of concern when designing an isolation system. Transmissibility is a term used to express the efficiency of the

isolation system. It is a ratio of forces. For example: If an unbalanced motor was running at a speed of 180 rpm and was producing 100 pounds of centrifugal force and it was desired to limit the force to 2 pounds on the foundation that the machine was set in, the transmissibility would be 0.02 or 2/100.

The reduction of structural-borne noise by vibration isolation can also be expressed in terms of transmissibility. On a decibel scale, it is defined as

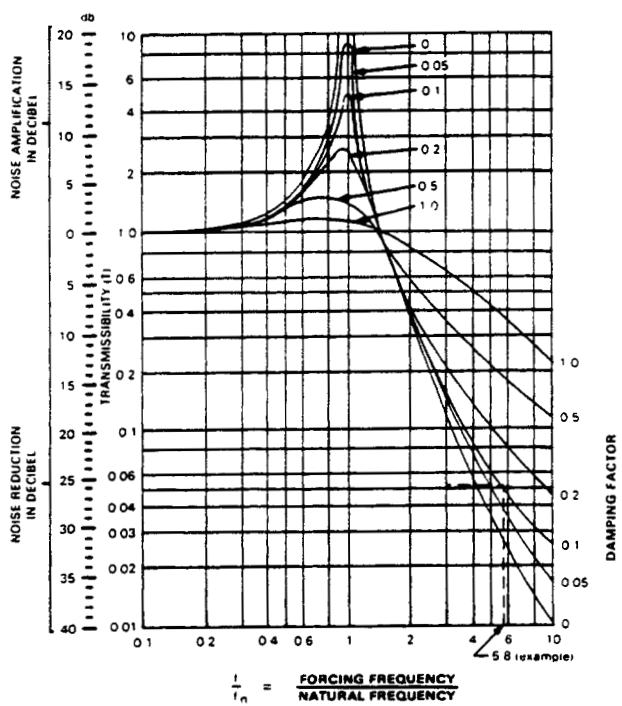
$$\text{Noise Reduction (NR in dB)} = 20 \text{ Log } T$$

$$= 20 \text{ Log } \frac{\text{Exciting force}}{\text{Transmitted force}}$$

Figure 4.5.2 is an illustration of transmissibility curves. The left-hand side of the figure indicates that if a noise were at 30 dB and it was desired to reduce it to 25 dB, the transmissibility of the system would have to be 0.6.

Figure 4.5.2

Transmissibility curves.



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The figure not only shows the relationship between transmissibility and the decibel scales but also gives transmissibility curves over various ratios of forcing frequency to natural frequency. On the horizontal axis is the ratio of forcing frequency to the natural frequency, and on the vertical axis is the transmissibility. Since low transmissibilities are essential for effective vibration isolation, it is only necessary to examine the transmissibility curves to find the desirable frequency ratio (forcing frequency to natural frequency) which must be attained to make the system effective. Notice from the curves that a reduction in noise in decibels occurs when the transmissibility is less than 1. If the transmissibility is greater than 1, there is an amplification in the noise. Other facts can be observed from the curves.

1. When the forcing frequency equals the natural frequency of the mounted system, the transmissibility becomes much greater than 1.0, and no isolation is attained. Instead, an amplification of the existing force or motion occurs (resonance).

2. It is not until the forcing frequency is 1.4 or more times the natural frequency of the mounted system that any isolation is achieved.

3. To achieve a transmissibility of 0.1, which means 90% isolation efficiency, the frequency ratio must be at least 3.3.

The above paragraphs suggest a step-by-step procedure for selecting a vibration isolator. These steps are:

1. Determine the lowest forcing frequency in the machine to be isolated.

2. Establish permissible transmissibility or desirable noise reduction level.

3. Using Figure 4.5.2, determine the required natural frequency.

4. Determine the necessary deflection to obtain the natural frequency required. Also consider the effect of the environmental conditions, such as creep, wear, resistance to corrosion, ozone, oil, or other detrimental agents.

5. Determine the weight of each mounting point on the machine to be isolated.

6. From the load deflection data supplied by the manufacturer of the vibration isolators, determine the size required.

A 200-pound machine is transmitting vibration at 40 Hz into a support structure with the force of 100 pounds. It is desired to limit the force to 5 pounds, by the use of an isolator. This would require a transmissibility of 0.05 (5/100). The machine is exposed to grease; hence, an isolator made of neoprene is desired. The dampening factor of the isolator is 0.1 according to the vendor. Following the 0.1 curve on Figure 4.5.2 to the transmissibility

of 0.05, the ratio of the forcing frequency to natural frequency is found to be as follows:

$$f/f_n = 5.8$$

Thus,

$$f_n = 40 \text{ Hz}/5.8 = 6.9 \text{ Hz}$$

From this information it can be seen that the static deflection of about 0.21 inches is required for the isolator. That is, the natural frequency of 6.9 Hz can be substituted in the following expression (assume load deflection curve is linear):

$$f_n = 3.13 \times 1/\sqrt{S_{st}}$$

Assuming equal distribution of load on the four points of support, the load per mount becomes 50 pounds, so that the stiffness of the isolator must be 50 lbs/0.21 inches, or 238 pounds per inch. From the information in the vendor's catalog, it is now possible to select the proper isolator.

Here is another example. If the vibration of a 200-pound machine is causing the supporting structure to radiate a 60-decibel noise, and it is desired to reduce the noise to 34 dB--a reduction of 26 decibels--the same procedure is used. First, follow the 0.1 curve on Figure 4.5.2 to the noise reduction of 26 dB, and the solution is obtained.

There are many types of isolators. Felt compositions called elastomers are effective at frequencies of about 30 Hz. There are spring elastomers which are not affected by oil or other contaminants. The type of isolator selected depends upon the particular problem to be solved.

The third method of control is reducing the response of the vibrating surface. This method would be particularly useful when the vibrating force is equal to the natural frequency of the vibrating surface, and resonance results. There are three methods of reducing the response of the vibrating surface: stiffening, increasing mass, and dampening. It should be noted that any of these methods would change the natural frequency of the system.

Usually large, flat surfaces will vibrate, and adding stiffeners will reduce the vibration. Along with the stiffening of a material, the shape of the vibrating surface can be changed. For example, the panel can be dished or curved so that the vibration is cut down. Also, the size of the radiating surface can be changed. A smaller surface will tend to radiate less sound.

The second method of reducing the response of the vibrating surface is through increasing the mass of the material or vibrating surface. Increased mass makes the material heavier and more resistant to vibration.However, when increasing the mass of a material, one must be careful of the resonant frequency of the new material.

The third method of reducing the response of the vibrating surface is through dampening. Dampening refers to restraining noise achieved by reducing

the contact of the vibrating surface with the air. This contact can be reduced by the addition of an absorbing material to the vibrating surface. This absorbing material will transfer the vibration energy into heat energy, and less energy will be put into the air. Dampening materials include dampening felt, elastomers, tars, and any kind of masking material. Optimizing dampening treatment is usually a complicated procedure. In addition, it can be extremely expensive. Thus, it is suggested than a technical expert should be consulted when considering dampening.

Summary

The following are key items from this chapter.

1. Vibration can be periodic or nonperiodic.

2. Vibration can be measured by examining displacement, velocity, and acceleration. The quantity used depends upon the vibration and the vibrating source.

3. Vibration measurements involve:

   a. determining what quantity to measure.
   b. determining the location to measure.
   c. determining how to fasten the vibration measuring instrument.

4. Vibration measurements are usually taken in several locations; e.g., perpendicular to shaft or housing.

5. Vibration measurement instruments include:

   a. pickups
   b. amplifiers
   c. analyzers

6. Other vibration measurement techniques include stroboscopic examination.

7. Vibration not only causes airborne noise but also causes whole-body and segmental vibration effects. These effects can cause physiological and psychological problems.

8. No reliable criteria exist for determining the vibration effects in an industrial setting.

9. Vibration can be controlled by:

   a. reducing the mechanical driving force
   b. isolating the vibration driving force from the radiating surface
   c. reducing the radiation of the vibrating surface

10. Obtaining vibration measurements and controlling vibration depends to a large extent on experience.